

VAIBHAV KARBHANTNAL

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SUMMARY

MCA graduate with a strong foundation in backend engineering, distributed systems, and applied AI/ML. Proven ability to architect and deploy production-grade systems — from event-driven microservices on AWS to real-time geospatial ML pipelines. Experienced in full-stack development, cloud infrastructure (Docker, Nginx, EC2), and building reliable, observable systems with tools like Datadog and Sentry. Proven expertise in applied ML and Agentic AI, deploying RAG pipelines and optimizing NLP inference endpoints for production.

SKILLS

Languages	Python, JavaScript, C++, Java, SQL, HTML/CSS
Backend & UI	FastAPI, Flask, Node.js, Next.js, React, Tailwind, RESTful APIs
DevOps & Cloud	Docker, Docker Compose, AWS (EC2), Azure, Nginx, Linux, Vercel
Databases & Brokers	MySQL, MongoDB, Redis, Celery, SQLAlchemy, SQLite
AI, ML & Data	Hugging Face, LangChain, Scikit-learn, XGBoost, Pandas, MLflow, NLP
Architecture	Microservices, Event-Driven, Cache-Aside, Serverless, RAG Pipelines
Security & Obs.	Redis ZSET Rate Limiting, Datadog APM, Sentry, JWT/Bearer Auth

INTERNSHIP

Backend Developer Intern

May 2025 – June 2025

RoborosX Omni Tech Solutions LLP

Bangalore, Karnataka

Skills/Tools used: Python, Flask, MySQL, REST API, SQLAlchemy, Bearer Token Auth, Git/GitHub

Contribution:

- Engineered RESTful APIs and centralized backend services for **RosX Edu Soft**, an enterprise ERP platform streamlining internal operations across the corporate network.
- Designed secure authentication pipelines using Bearer Token authorization, hardening access control for sensitive internal endpoints.
- Optimized database queries and schema design to support scalable, high-performance data retrieval across distributed organizational services.

Subject Matter Expert (SME)

June 2023 – November 2023

Chegg (Direction Educare)

Punjab, India

Contribution:

- Delivered comprehensive technical solutions to complex computer science problems, ensuring high accuracy and rigorous analytical depth across a large volume of academic queries.

PROJECTS

Event-Driven Mailer Platform (V2.0)

Live Production

mailer-system.vaibhavkarbhantnal.me | github.com/NukGod-V/Mailer-System

- Architected a decoupled microservice utilizing a **Redis** broker and **Celery** background workers to asynchronously dispatch email payloads, fully eliminating UI blocking.
- Containerized via **Docker Compose** and deployed to **AWS EC2** behind an **Nginx** reverse proxy with automated SSL. Instrumented with **Datadog APM** and **Sentry** for real-time observability.
- Engineered an in-memory Base64 1x1 tracking pixel and a dual-layer time-math algorithm to filter out enterprise security bots (e.g., GoogleImageProxy), achieving **100% accurate human open-rate analytics**.

Tech Stack: Python, Flask, Redis, Celery, AWS EC2, Datadog APM, Nginx, Sentry.

Real-Time Forest Fire Risk Prediction

Live Serverless Deployment

forest-fire.vaibhavkarbhantnal.me | github.com/NukGod-V/Forest-Fire-Monitor

- Architected an asynchronous backend to ingest near real-time thermal anomaly data from the **NASA FIRMS API**, decoupling the ingestion logic into **Microsoft Azure Functions** to lower idle compute costs.
- Trained and served an **XGBoost** model (99% accuracy) and built a geospatial **Streamlit/Folium** dashboard.

Tech Stack: Python, Azure Functions, FastAPI, XGBoost, Streamlit, Folium, NASA FIRMS API.

Agentic RAG Portfolio Assistant (Digital Clone)

Month 4 Capstone

vaibhavkarbhantnal.me | github.com/NukGod-V/Personal-LLM

- Architected a customized **LangChain** RAG pipeline deployed via **Camber Cloud** to serve as an interactive, context-aware AI portfolio assistant.
- Developed a secure **Next.js/React** frontend chat interface to query the Camber Cloud NLP endpoint, utilizing serverless API routes (BFF) to strictly mask credentials and manage conversational state.

Tech Stack: Next.js, React, Tailwind CSS, LangChain, Camber Cloud, RAG, Prompt Engineering.

High-Concurrency ML Inference API & Rate Limiter

Production ML

github.com/NukGod-V

- Deployed a high-concurrency Sentiment Analysis API using **FastAPI** and **Hugging Face/Scikit-Learn**, handling real-world NLP inputs with 88% F1-score accuracy and preserving semantic negation context.
- Integrated **MLflow** to manage the model lifecycle, establishing an experiment registry for automated hyperparameter tracking and preventing training-serving skew.
- Engineered a highly optimized Sliding Window Rate Limiter using **Redis Sorted Sets (ZSET)** and atomic pipelines to enforce strict IP-based traffic thresholds and protect endpoints from DoS loads.

Tech Stack: Python, FastAPI, Redis, MLflow, Hugging Face, Scikit-Learn, Docker, NLP.

ACHIEVEMENTS & CONTINUOUS LEARNING

- Awarded **Student of the Year 2023** in the BCA program for academic excellence and significant contributions to university projects and community initiatives.
- **Algorithmic Optimization:** Actively progressing through the **NeetCode 150** framework to continuously refine data structure mastery, runtime complexity analysis, and problem-solving methodology.

EDUCATION

BMS Institute of Technology and Management, Bangalore, Karnataka

Master of Computer Applications (MCA)

February 2024 – February 2026

Graduation CGPA: 8.47/10

Rani Channamma University (A.S. Patil), Bijapur, Karnataka

Bachelor of Computer Applications (BCA)

September 2020 – October 2023

Graduation CGPA: 8.73/10

Roopadevi PU Science College, Bijapur, Karnataka

Pre-University Course (PUC) in Science

June 2018 – July 2020

Percentage: 68%

Prerana Public School (CBSE), Bijapur, Karnataka

Secondary School Leaving Certificate (SSLC)

July 2018